

## Questions with Answer Keys

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## Q1 - 2024 (01 Feb Shift 1)

A plane is in level flight at constant speed and each of its two wings has an area of  $40 \text{ m}^2$ . If the speed of the air is  $180 \text{ km/h}$  over the lower wing surface and  $252 \text{ km/h}$  over the upper wing surface, the mass of the plane is \_\_\_\_\_ kg. (Take air density to be  $1 \text{ kg m}^{-3}$  and  $g = 10 \text{ ms}^{-2}$ )

## Q2 - 2024 (01 Feb Shift 2)

A big drop is formed by coalescing 1000 small droplets of water. The surface energy will become :

(1) 100 times

(2) 10 times

(3)  $\frac{1}{100}$  th

(4)  $\frac{1}{10}$  th

## Q3 - 2024 (27 Jan Shift 1)

Given below are two statements :

Statement (I) : Viscosity of gases is greater than that of liquids.

Statement (II) : Surface tension of a liquid decreases due to the presence of insoluble impurities.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are incorrect

(4) Both Statement I and Statement II are correct

## Q4 - 2024 (27 Jan Shift 2)

The reading of pressure metre attached with a closed pipe is  $4.5 \times 10^4 \text{ N/m}^2$ . On opening the valve, water starts flowing and the reading of pressure metre falls to  $2.0 \times 10^4 \text{ N/m}^2$ . The velocity of water is found to be  $\sqrt{V} \text{ m/s}$ . The value of V is \_\_\_\_\_

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## Q5 - 2024 (29 Jan Shift 1)

Given below are two statements:

Statement I : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in hot water.

Statement II : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in cold water.

In the light of the above statements, choose the most appropriate from the options given below

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

## Q6 - 2024 (29 Jan Shift 1)

In a test experiment on a model aeroplane in wind tunnel, the flow speeds on the upper and lower surfaces of the wings are  $70 \text{ ms}^{-1}$  and  $65 \text{ ms}^{-1}$  respectively. If the wing area is  $2 \text{ m}^2$  the lift of the wing is \_\_\_\_\_ N.  
(Given density of air =  $1.2 \text{ kg m}^{-3}$ )

## Q7 - 2024 (29 Jan Shift 2)

A small liquid drop of radius  $R$  is divided into 27 identical liquid drops. If the surface tension is  $T$ , then the work done in the process will be :

(1)  $8\pi R^2 T$

(2)  $3\pi R^2 T$

(3)  $\frac{1}{8}\pi R^2 T$

(4)  $4\pi R^2 T$

## Q8 - 2024 (30 Jan Shift 2)

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## Questions with Answer Keys

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A big drop is formed by coalescing 1000 small identical drops of water. If  $E_1$  be the total surface energy of 1000 small drops of water and  $E_2$  be the surface energy of single big drop of water, the  $E_1 : E_2$  is  $x : 1$  where  $x =$  \_\_\_\_\_

## Q9 - 2024 (31 Jan Shift 2)

A small spherical ball of radius  $r$ , falling through a viscous medium of negligible density has terminal velocity ' $v$ '. Another ball of the same mass but of radius  $2r$ , falling through the same viscous medium will have terminal velocity:

- (1)  $\frac{v}{2}$
- (2)  $\frac{v}{4}$
- (3)  $4v$
- (4)  $2v$

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**Answer Key**

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## Solutions

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Q1

$$A = 80 \text{ m}^2$$

Using Bernonlli equation

$$A(P_2 - P_1) = \frac{1}{2}\rho(V_1^2 - V_2^2)A$$

$$mg = \frac{1}{2} \times 1(70^2 - 50^2) \times 80$$

$$mg = 40 \times 2400$$

$$m = 9600 \text{ kg}$$

Q2

Lets say radius of small droplets is r and that of big drop is R

$$\frac{4}{3}\pi R^3 = 1000 \frac{4}{3}\pi r^3$$

$$R = 10r$$

$$U_i = 1000(4\pi r^2 S)$$

$$U_f = 4\pi R^2 S$$

$$= 100(4\pi r^2 S)$$

$$U_f = \frac{1}{10}U_i$$

Q3

Gases have less viscosity.

Due to insoluble impurities like detergent surface tension decreases

Q4

$$\text{Change in pressure} = \frac{1}{2}\rho v^2$$

$$4.5 \times 10^4 - 2.0 \times 10^4 = \frac{1}{2} \times 10^3 \times v^2$$

$$2.5 \times 10^4 = \frac{1}{2} \times 10^3 \times v^2$$

$$v^2 = 50$$

$$v = \sqrt{50}$$

$$\text{Velocity of water} = \sqrt{V} = \sqrt{50}$$

$$= V = 50$$

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## Solutions

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Q5

Surface tension will be less as temperature increases

$$h = \frac{2 T \cos \theta}{\rho g r}$$

Height of capillary rise will be smaller in hot water and larger in cold water.

Q6

$$F = \frac{1}{2} \rho (v_1^2 - v_2^2) A$$

$$F = \frac{1}{2} \times 1.2 \times (70^2 - 65^2) \times 2$$

$$= 810 \text{ N}$$

Q7

Volume constant

$$\frac{4}{3} \pi R^3 = 27 \times \frac{4}{3} \times \pi r^3$$

$$R^3 = 27 r^3$$

$$R = 3r$$

$$r = \frac{R}{3}$$

$$r^2 = \frac{R^2}{9}$$

$$\text{Work done} = T \cdot \Delta A$$

$$= 27 T (4\pi r^2) - T 4\pi R^2$$

$$= 27 T 4\pi \frac{R^2}{9} - 4\pi R^2 T$$

$$= 8\pi R^2 T$$

Q8

$$\rho \left( \frac{4}{3} \pi r^3 \right) 1000 = \frac{4}{3} \pi R^3 \rho$$

$$R = 10r$$

$$E_1 = 1000 \times 4\pi r^2 \times S$$

$$E_2 = 4\pi (10r)^2 S$$

$$\frac{E_1}{E_2} = \frac{10}{1}, x = 10$$

Q9

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## Solutions

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Since density is negligible hence Buoyancy force will be negligible

At terminal velocity.

$$Mg = 6\pi\eta rv$$

$$V \propto \frac{1}{r} \quad (\text{as mass is constant})$$

$$\text{Now, } \frac{v}{v'} = \frac{r'}{r}$$

$$r' = 2r$$

$$\text{So, } v' = \frac{v}{2}$$

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